NLP and Word Embeddings

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Word representation

Word representation

V = [a, aaron, …, zulu, <UNK>]

$|V| = 10{,}000$

1-hot representation

Man	Woman	King	Queen	Apple	Orange
(5391)	(9853)	(4914)	(7157)	(456)	(6257)

$$\begin{bmatrix}0\\0\\0\\0\\\vdots\\1\\\vdots\\0\\0\end{bmatrix} \quad \begin{bmatrix}0\\0\\0\\0\\0\\\vdots\\1\\\vdots\\0\end{bmatrix} \quad \begin{bmatrix}0\\0\\0\\\vdots\\1\\\vdots\\0\\0\\0\end{bmatrix} \quad \begin{bmatrix}0\\0\\0\\0\\0\\\vdots\\1\\\vdots\\0\end{bmatrix} \quad \begin{bmatrix}0\\\vdots\\1\\\vdots\\0\\0\\0\\0\\0\end{bmatrix} \quad \begin{bmatrix}0\\0\\0\\0\\0\\\vdots\\1\\\vdots\\0\end{bmatrix}$$

O_{5391} O_{9853}

I want a glass of orange juice.

I want a glass of apple ?.

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Featurized representation: word embedding

	Man (5391)	Woman (9853)	King (4914)	Queen (7157)	Apple (456)	Orange (6257)
Gender	-1	1	-0.95	0.97	0.00	0.01
Royal	0.01	0.02	0.93	0.95	-0.01	0.00
Age	0.03	0.02	0.7	0.69	0.03	-0.02
Food	0.04	0.01	0.02	0.01	0.95	0.97
⋮	⋮	⋮				
size						
cost						
alive						
verb						
	e_{5391}	e_{9853}				

300

I want a glass of orange juice.

I want a glass of apple juice.

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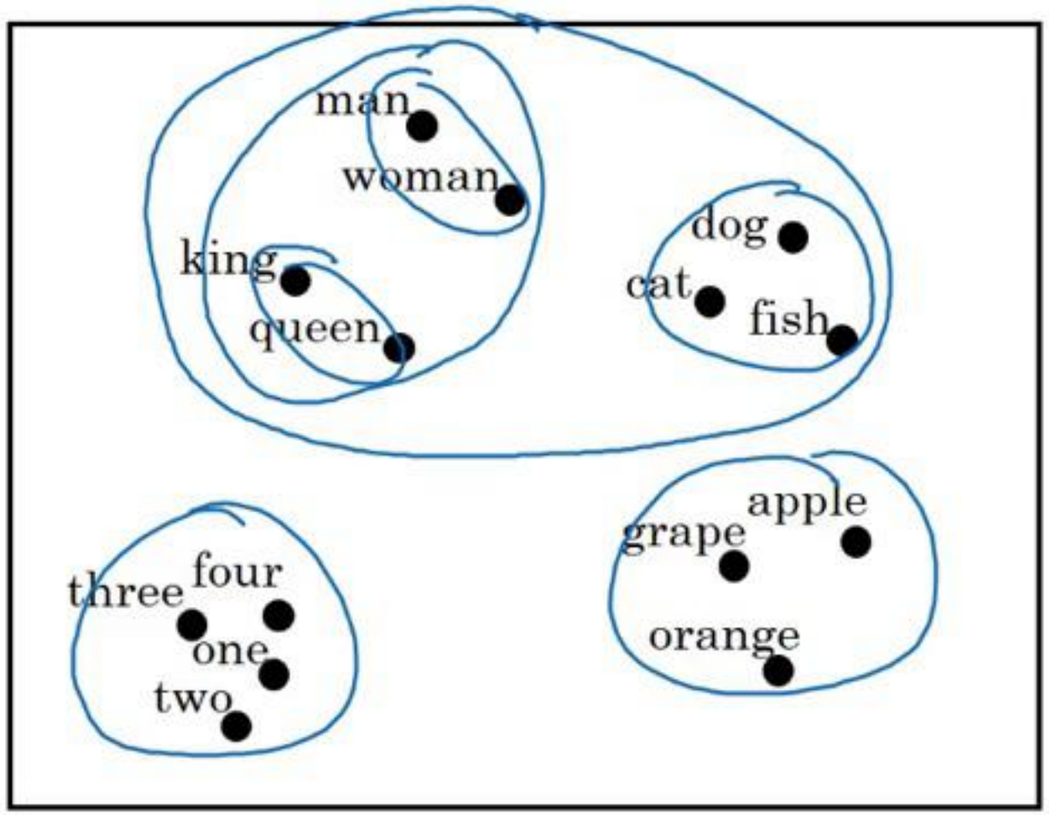
Visualizing word embeddings

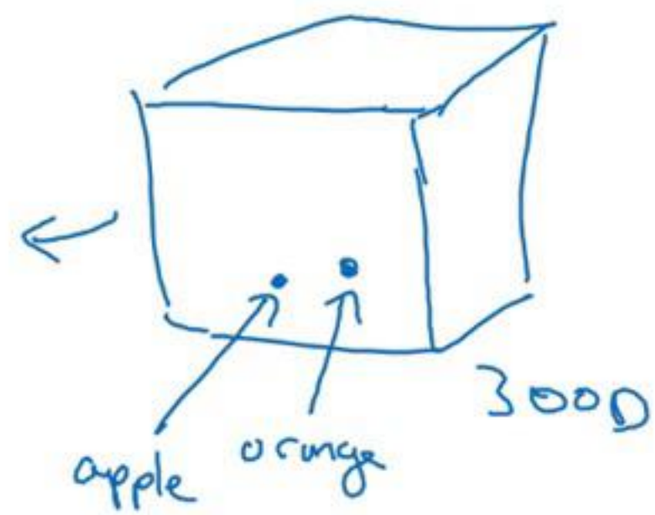


t-SNE

→ 300D

↓

2D



[van der Maaten and Hinton., 2008. Visualizing data using t-SNE]

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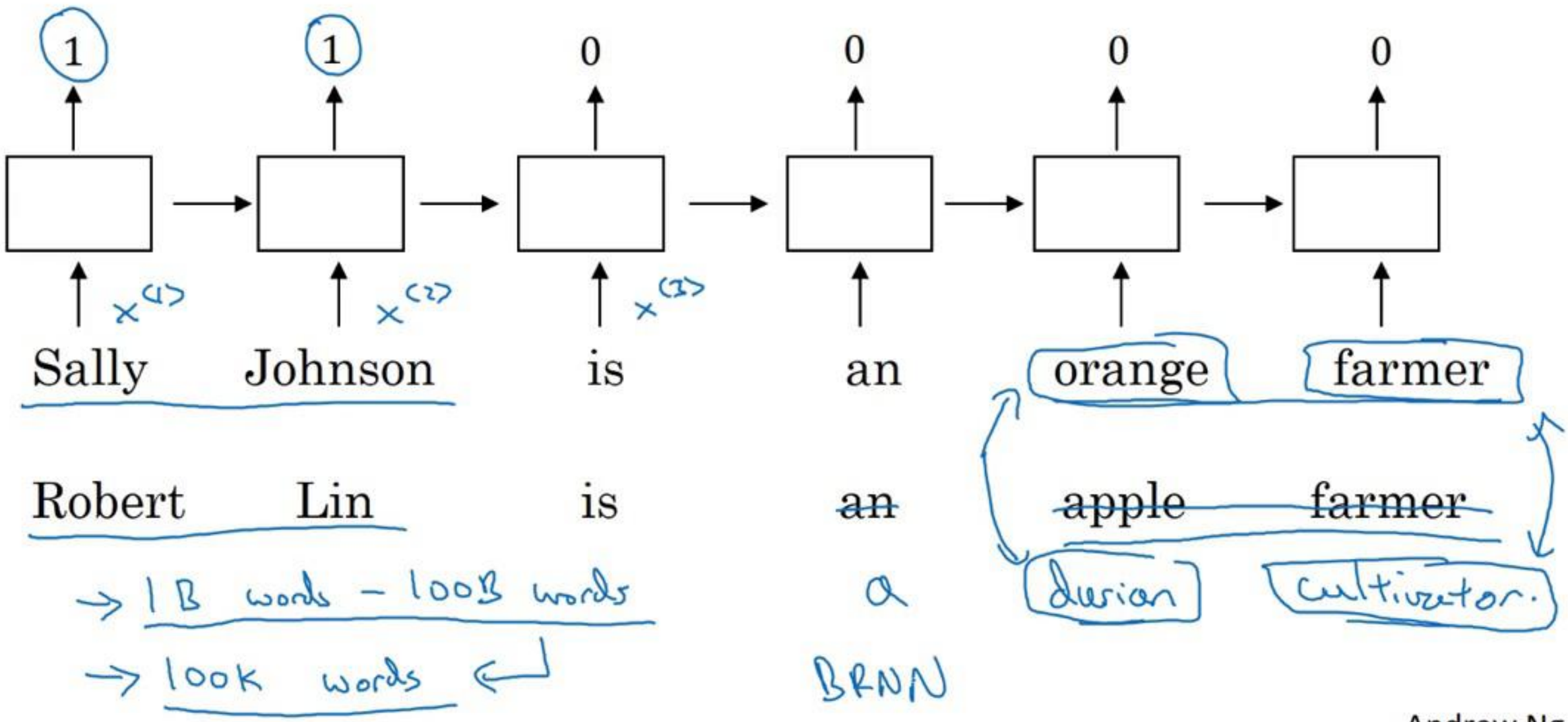
Using word embeddings

Named entity recognition example



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Transfer learning and word embeddings

A

1. Learn word embeddings from large text corpus. (1-100B words)

   (Or download pre-trained embedding online.)

B

2. Transfer embedding to new task with smaller training set. (say, 100k words) → 10,000 → 300

3. Optional: Continue to finetune the word embeddings with new data.

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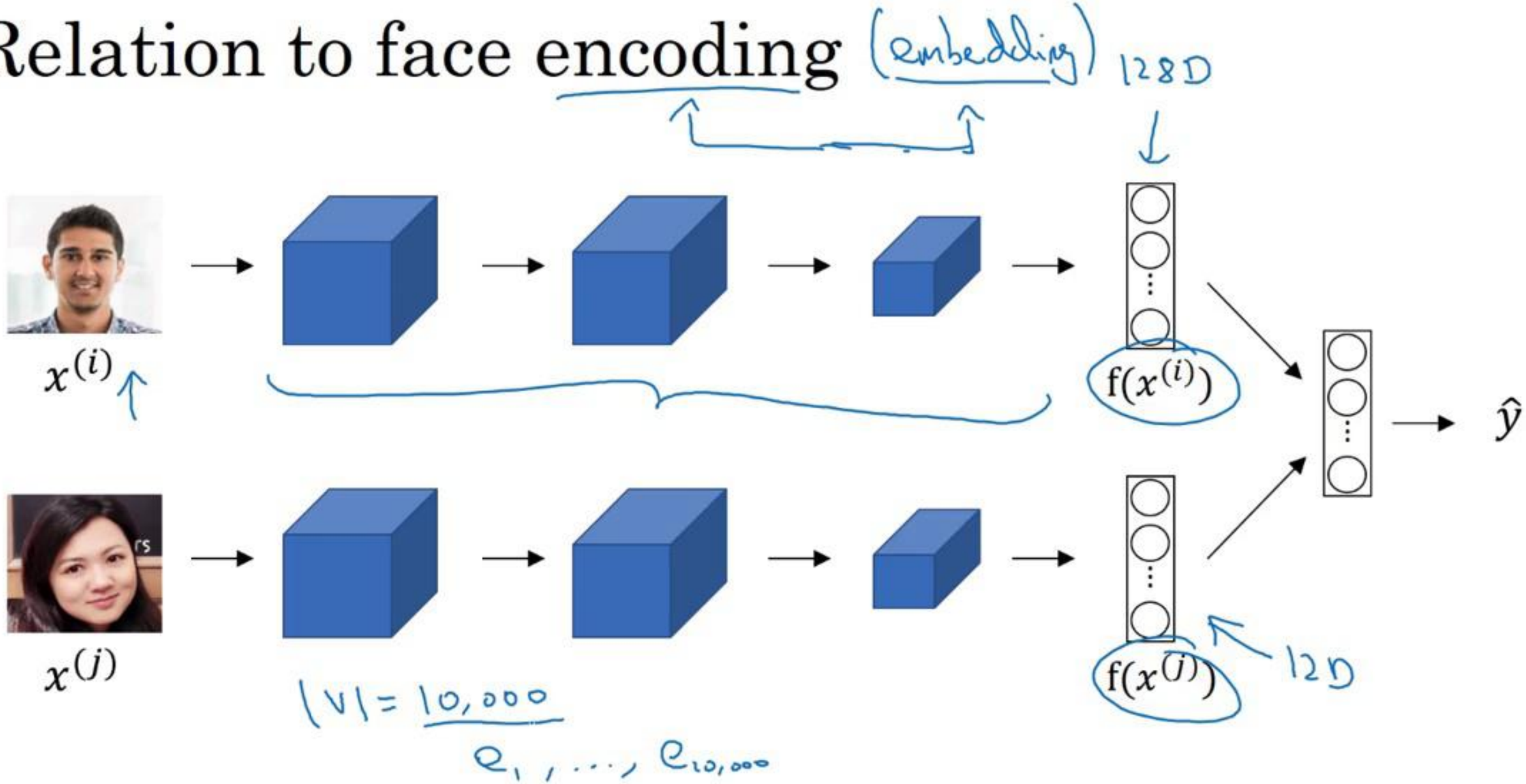
Relation to face encoding (embedding)



[Taigman et. al., 2014. DeepFace: Closing the gap to human level performance]

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Properties of word embeddings

Analogies

	Man (5391)	Woman (9853)	King (4914)	Queen (7157)	Apple (456)	Orange (6257)
Gender	−1	1	-0.95	0.97	0.00	0.01
Royal	0.01	0.02	0.93	0.95	-0.01	0.00
Age	0.03	0.02	0.70	0.69	0.03	-0.02
Food	0.09	0.01	0.02	0.01	0.95	0.97

e_{5391}
e_{man}

e_{woman}

$e_{man} - e_{woman} \approx \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Man → Woman as King → ? Queen

$e_{king} - e_{queen} \approx \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$e_{man} - e_{woman} \approx e_{king} - e_{?}$

Queen

[Mikolov et. al., 2013, Linguistic regularities in continuous space word representations]

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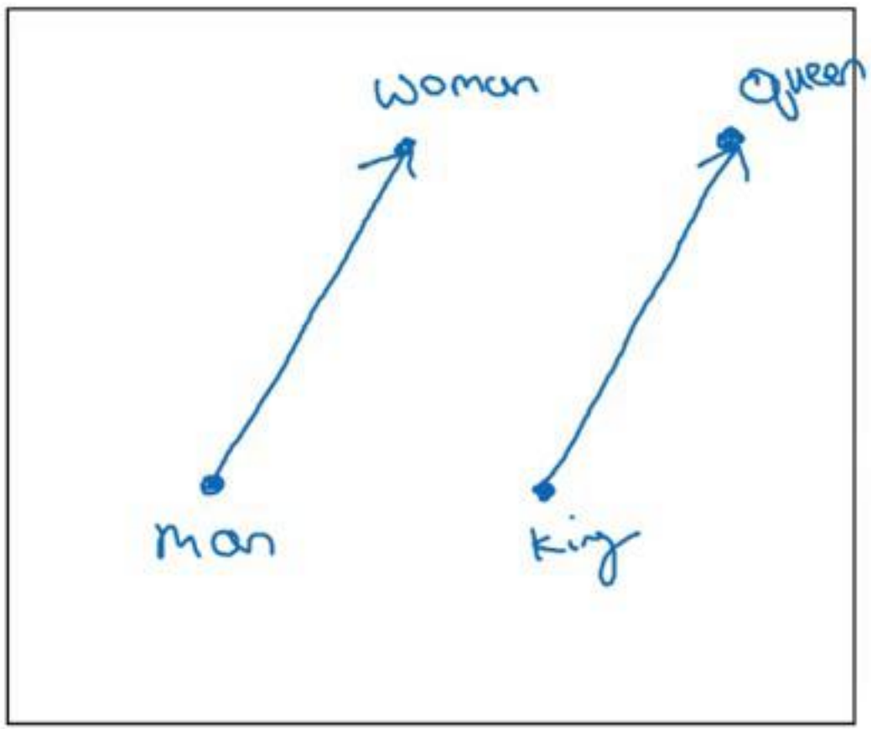
Analogies using word vectors



300 D

$$e_{man} - e_{woman} \approx e_{king} - \cancel{e_?}\ e_w$$

Find word w: $\arg\max_w \ \mathrm{sim}(e_w, e_{king} - e_{man} + e_{woman})$

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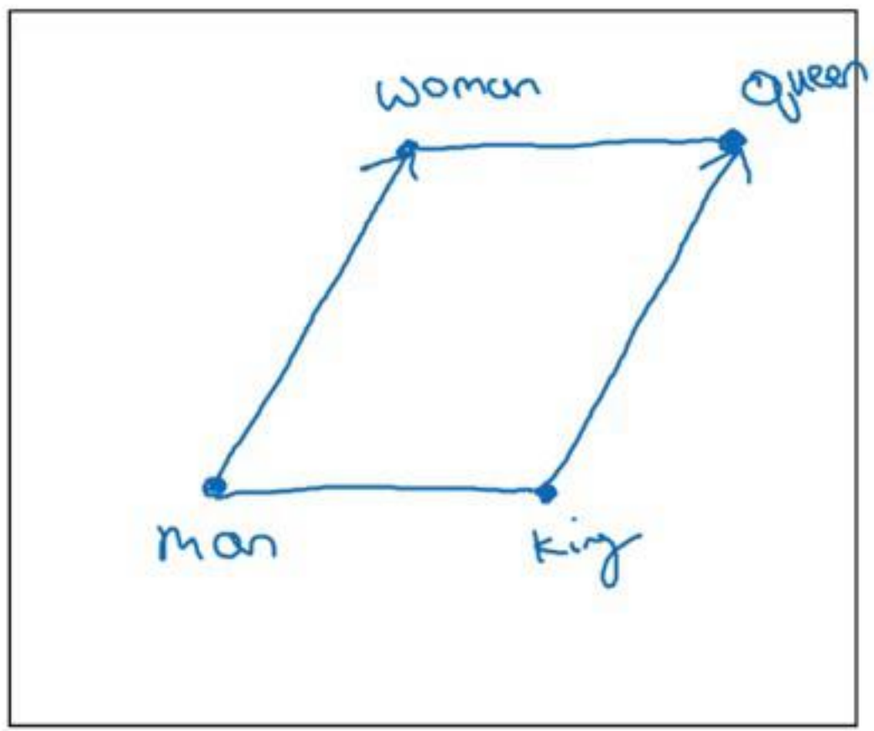
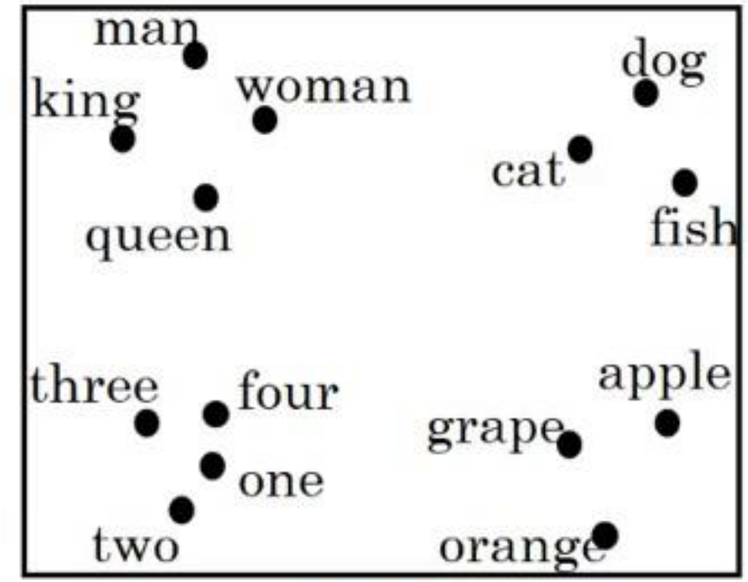
Analogies using word vectors



300 D

3000 → 20



t-SNE

$$e_{man} - e_{woman} \approx e_{king} - \cancel{e_?}\ e_w$$

Find word w: $\arg\max_w \boxed{\mathrm{sim}(e_w, e_{king} - e_{man} + e_{woman})}$

30 - 75%

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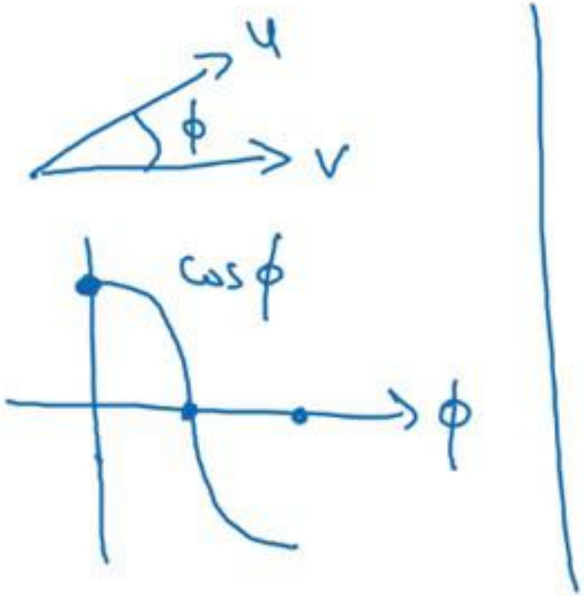
Cosine similarity

$$sim(e_w, e_{king} - e_{man} + e_{woman})$$

$$sim(u, v) = \frac{u^T v}{\|u\|_2 \|v\|_2}$$



$\|u - v\|^2$

Man:Woman as Boy:Girl

Ottawa:Canada as Nairobi:Kenya

Big:Bigger as Tall:Taller

Yen:Japan as Ruble:Russia

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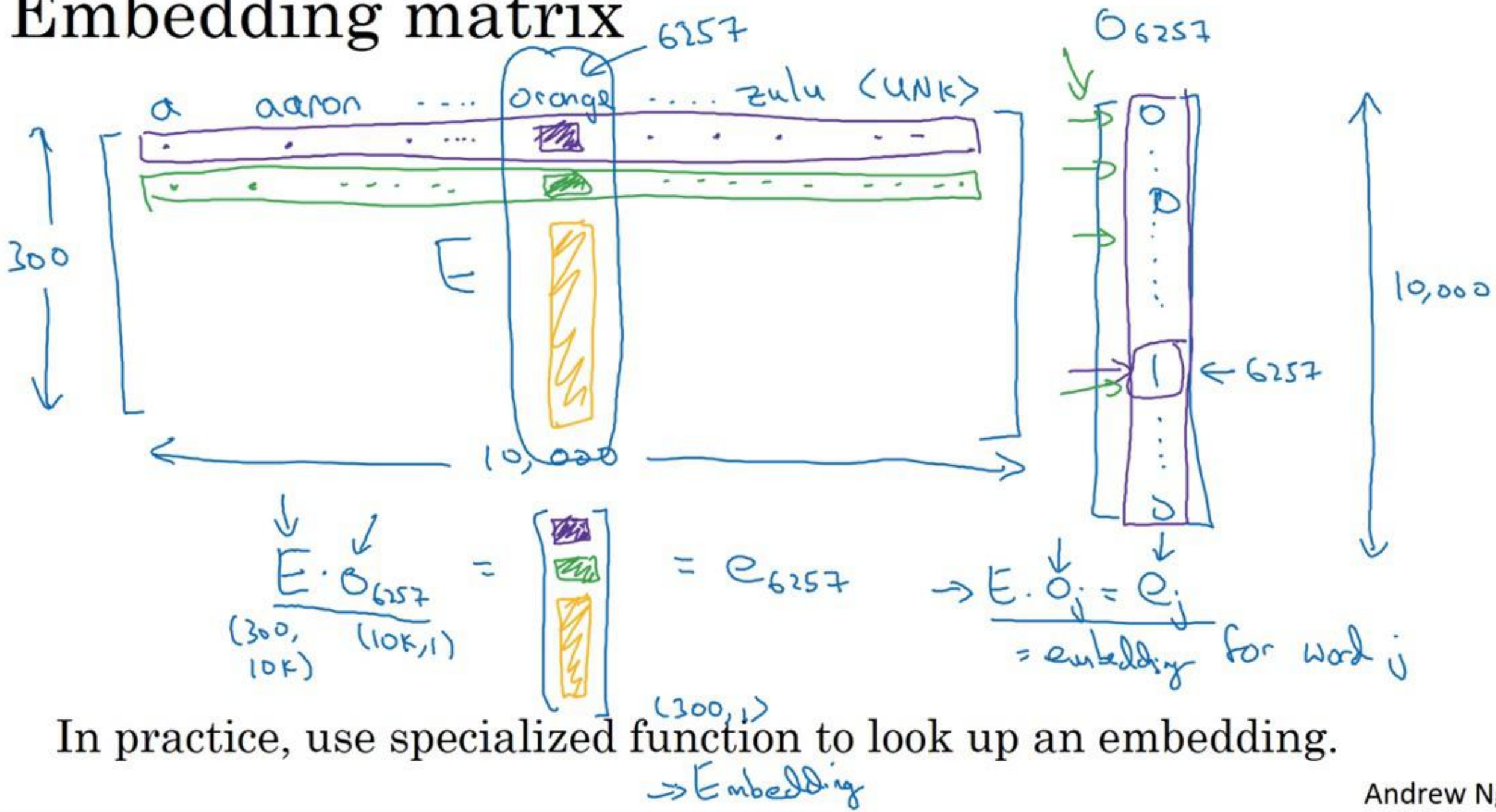
Embedding matrix

Embedding matrix



In practice, use specialized function to look up an embedding.

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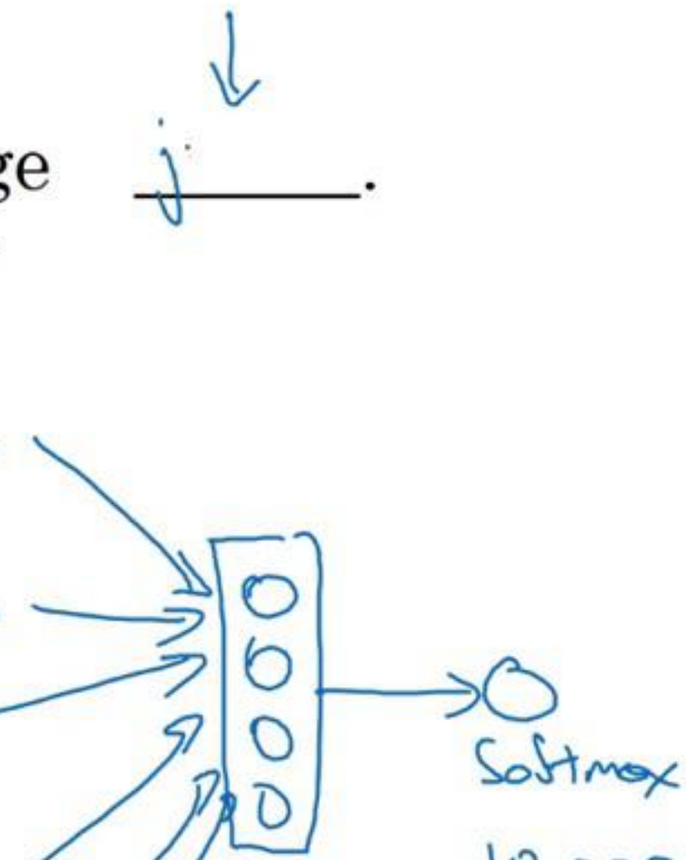
Learning word embeddings

Neural language model

I	want	a	glass	of	orange	____.
4343	9665	1	3852	6163	6257	

$e_{4343} = E\, o_{4343}$

I	o_{4343}	$\longrightarrow$	E	$\longrightarrow$ e_{4343}
want	o_{9665}	$\longrightarrow$	E	$\longrightarrow$ e_{9665}
a	o_1	$\longrightarrow$	E	$\longrightarrow$ e_1
glass	o_{3852}	$\longrightarrow$	E	$\longrightarrow$ e_{3852}
of	o_{6163}	$\longrightarrow$	E	$\longrightarrow$ e_{6163}
orange	o_{6257}	$\longrightarrow$	E	$\longrightarrow$ e_{6257}



[Bengio et. al., 2003, A neural probabilistic language model]

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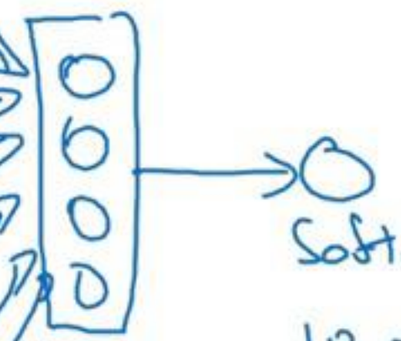
Neural language model

I	want	a	glass	of	orange	juice.
4343	9665	1	3852	6163	6257	

apple juice.

$e_{4343} = E\, o_{4343}$

I	o_{4343}	$\longrightarrow$	E	$\longrightarrow$	e_{4343}
want	o_{9665}	$\longrightarrow$	E	$\longrightarrow$	e_{9665}
a	o_1	$\longrightarrow$	E	$\longrightarrow$	e_1
glass	o_{3852}	$\longrightarrow$	E	$\longrightarrow$	e_{3852}
of	o_{6163}	$\longrightarrow$	E	$\longrightarrow$	e_{6163}
orange	o_{6257}	$\longrightarrow$	E	$\longrightarrow$	e_{6257}



Softmax

10,000

$W^{[1]}, b^{[1]}$

$W^{[2]}, b^{[2]}$

~~1800~~ 1200

[Bengio et. al., 2003, A neural probabilistic language model]

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Other context/target pairs

I want a glass of orange juice to go along with my cereal.

context

target

Context: Last 4 words.

- 4 words on left & right
- Last 1 word
- Nearby 1 word

a glass of orange ? to go along with

orange ?

glass ?

skip gram

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Word2Vec

Skip-grams

I want a glass of orange juice to go along with my cereal.

Context	Target
orange	juice
orange	glass
orange	my

[Mikolov et. al., 2013. Efficient estimation of word representations in vector space.]

Model

Vocab size = 10,000k

$x \longrightarrow y$

Context c ("orang") $\longrightarrow$ Target t ("juice")

6257 4834

$$o_c \rightarrow E \rightarrow e_c \rightarrow \underset{\text{softmax}}{O} \rightarrow \hat{y}$$

$$e_c = E o_c$$

Softmax: $$p(t|c) = \frac{e^{\theta_t^T e_c}}{\sum_{j=1}^{10,000} e^{\theta_j^T e_c}}$$

θ_t = parameter associated with output t

$$\rightarrow \mathcal{L}(\hat{y}, y) = -\sum_{i=1}^{10,000} y_i \log \hat{y}_i$$

$$y = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow 4834$$

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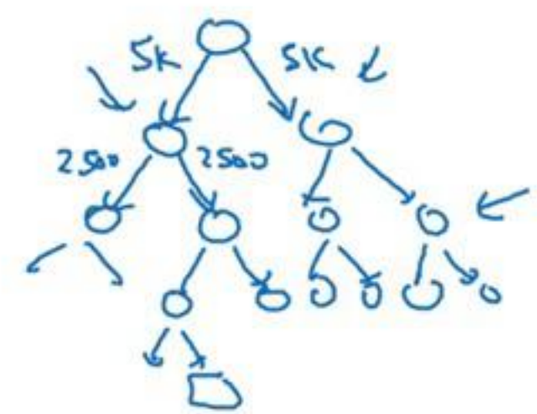
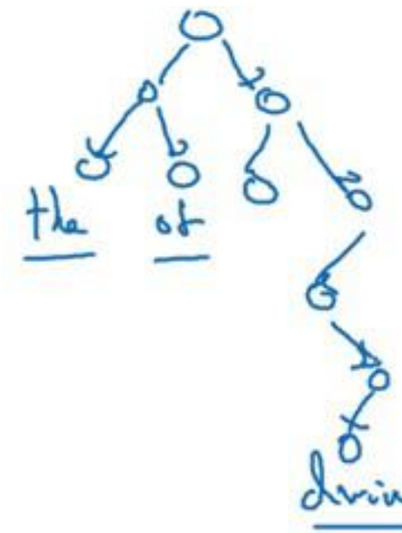
Problems with softmax classification

Hierarchical softmax.

$$p(t|c) = \frac{e^{\theta_t^T e_c}}{\boxed{\sum_{j=1}^{10,000} e^{\theta_j^T e_c}}}$$

$\log |V|$





Model

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Problems with softmax classification

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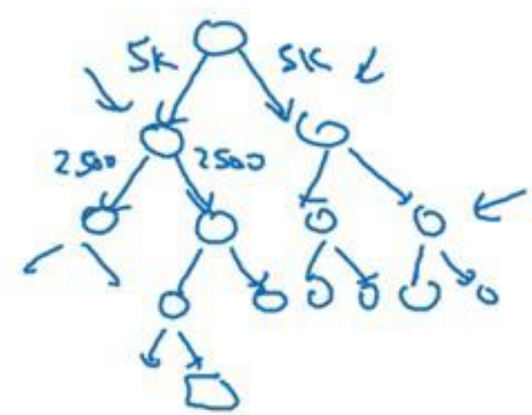
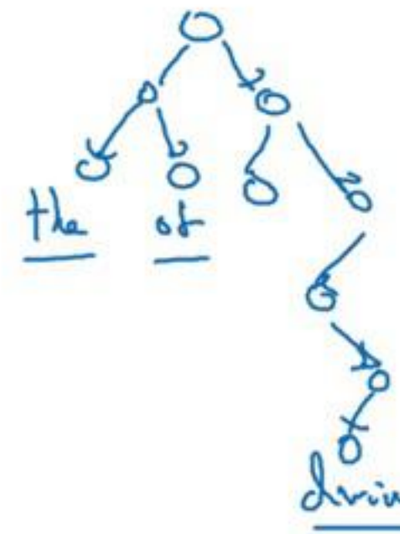
Hierarchical softmax.

log |V|





How to sample the context c?

t

$c \to t$

→ the, of, a, and, to, ...

→ orange, apple, durian

$P(c)$

e_{durian}

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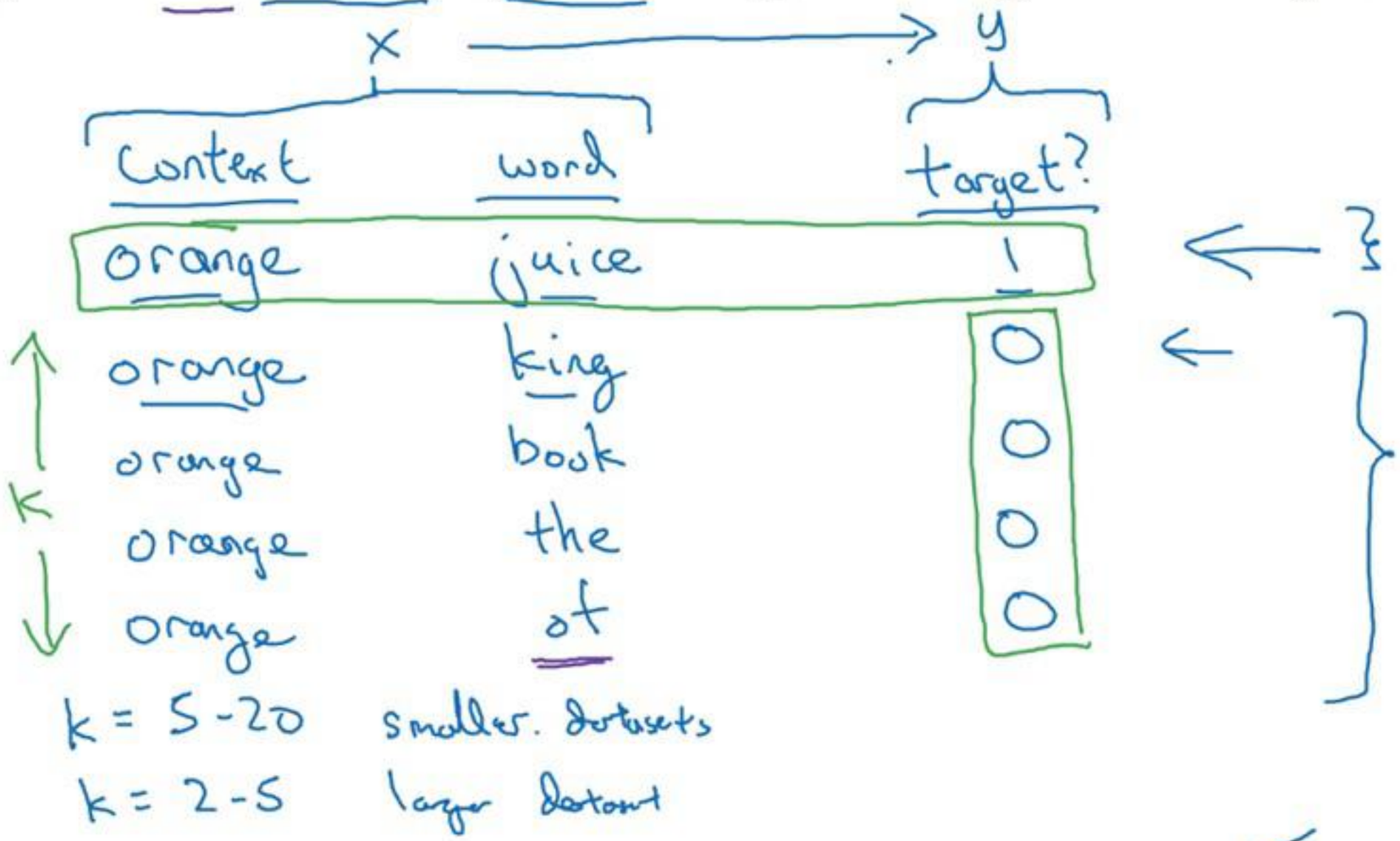
Negative sampling

Defining a new learning problem

I want a glass of orange juice to go along with my cereal.



[Mikolov et. al., 2013. Distributed representation of words and phrases and their compositionality]

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Model

Softmax: $$p(t|c) = \frac{e^{\theta_t^T e_c}}{\sum_{j=1}^{10,000} e^{\theta_j^T e_c}}$$

10,000-way softmax

$$P(y=1 \mid c, t) = \sigma(\theta_t^T e_c)$$

context	word	target?
orange	juice	1
orange	king	0
orange	book	0
orange	the	0
orange	of	0

x y

c t y

k

Orange 6257

$o_{6257} \rightarrow E \rightarrow e_{6257}$

juice?

king

10,000

10,000 binary classification problem

k+1

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Selecting negative examples

context	word	target?
orange	juice	1
orange	king	0
orange	book	0
orange	the	0
orange	of	0

the, of, and, ...

$$P(w_i) = \frac{f(w_i)^{3/4}}{\sum_{j=1}^{10,000} f(w_j)^{3/4}}$$

$$\frac{1}{|V|}$$

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GloVe word vectors

GloVe (global vectors for word representation)

I want a glass of orange juice to go along with my cereal.

c, t

X_{ij} = # times i appears in context of j.

c t t c

$X_{ij} = X_{ji}$

[Pennington et. al., 2014. GloVe: Global vectors for word representation]

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Model

$$\text{minimize} \quad \sum_{i=1}^{10{,}000} \sum_{j=1}^{10{,}000} \underbrace{f(X_{ij})}_{\text{weighting term}} \left(\theta_i^T e_j + b_i + b_j' - \log X_{ij} \right)^2$$

$\theta_i^T e_j$: i → t, j → c

b_i: i → t, b_j': j → c

X_{ij} → 0?

"$\theta_t^T e_c$"

$f(x_{ij}) = 0$ if $x_{ij} = 0$.

"$0 \log 0$" $= 0$

this, is, of, a, ...

durian

θ_i, e_j are symmetric

$$e_w^{(final)} = \frac{e_w + \theta_w}{2}$$

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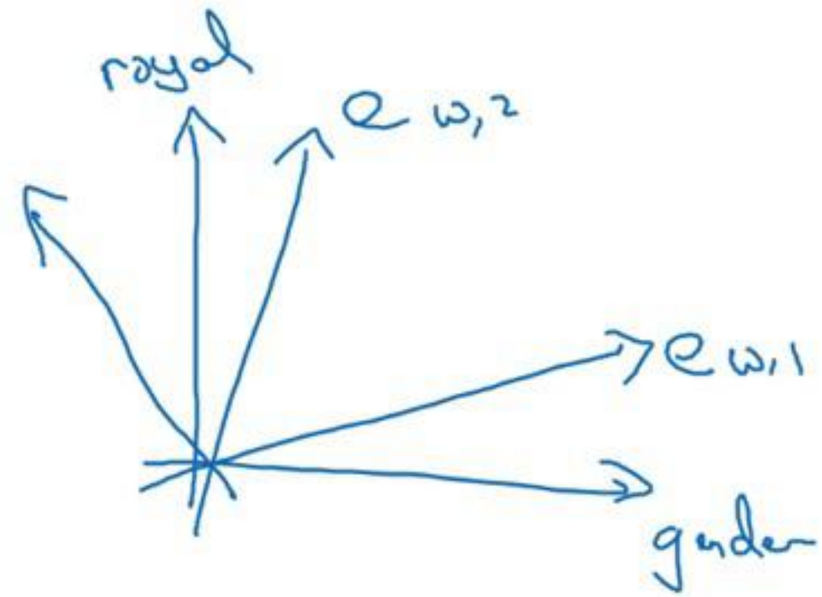
A note on the featurization view of word embeddings

	Man (5391)	Woman (9853)	King (4914)	Queen (7157)
Gender	−1	1	-0.95	0.97
Royal	0.01	0.02	0.93	0.95
Age	0.03	0.02	0.70	0.69
Food	0.09	0.01	0.02	0.01



$$\text{minimize} \sum_{i=1}^{10,000} \sum_{j=1}^{10,000} f(X_{ij})(\theta_i^T e_j + b_i - b_j' - \log X_{ij})^2$$

$$(A\theta_i)^T(A^{-T}e_j) = \theta_i^T \cancel{A^T A^{-T}} e_j$$

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Sentiment classification

Sentiment classification problem

$x \longrightarrow y$

The dessert is excellent.	★★★★☆
Service was quite slow.	★★☆☆☆
Good for a quick meal, but nothing special.	★★★☆☆
Completely lacking in good taste, good service, and good ambience.	★☆☆☆☆

10,000 → 100,000 words

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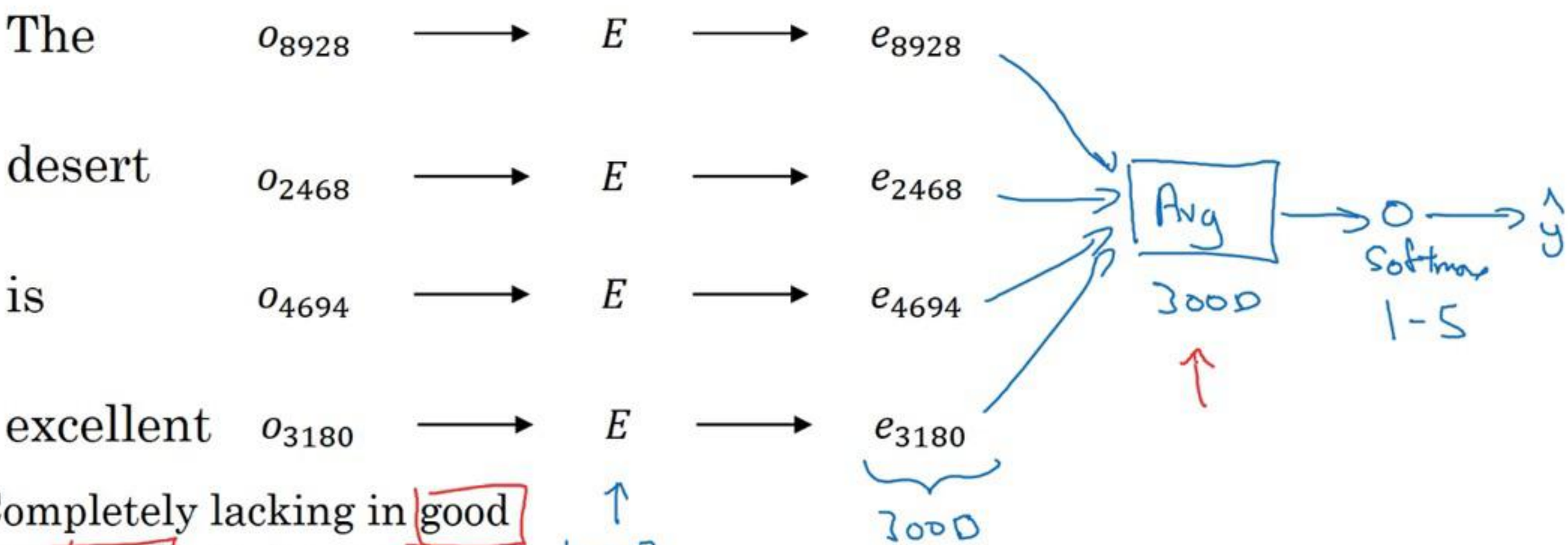
Simple sentiment classification model

The dessert is excellent ★★★★☆

8928 2468 4694 3180



"Completely lacking in good taste, good service, and good ambience."

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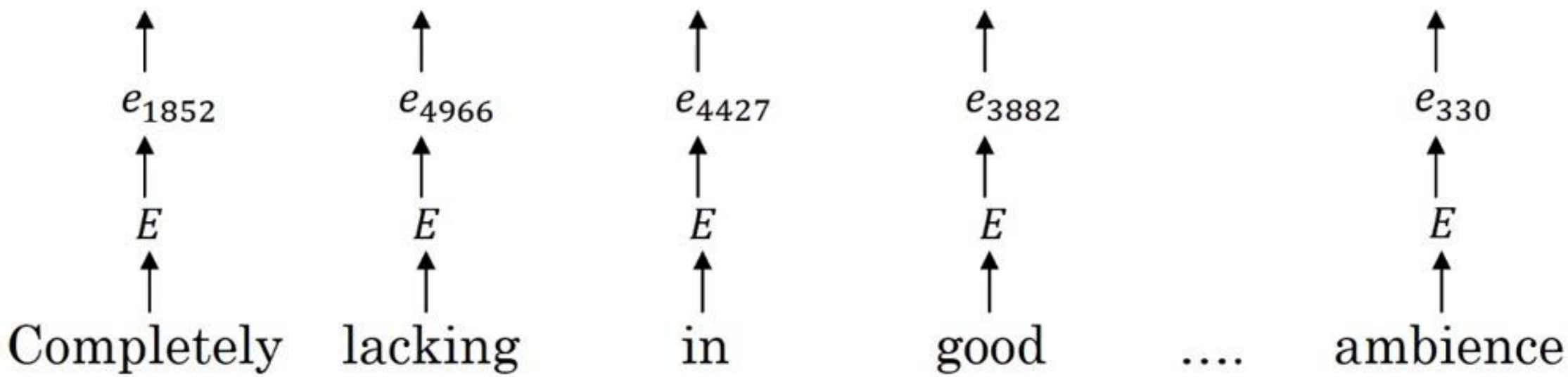
RNN for sentiment classification



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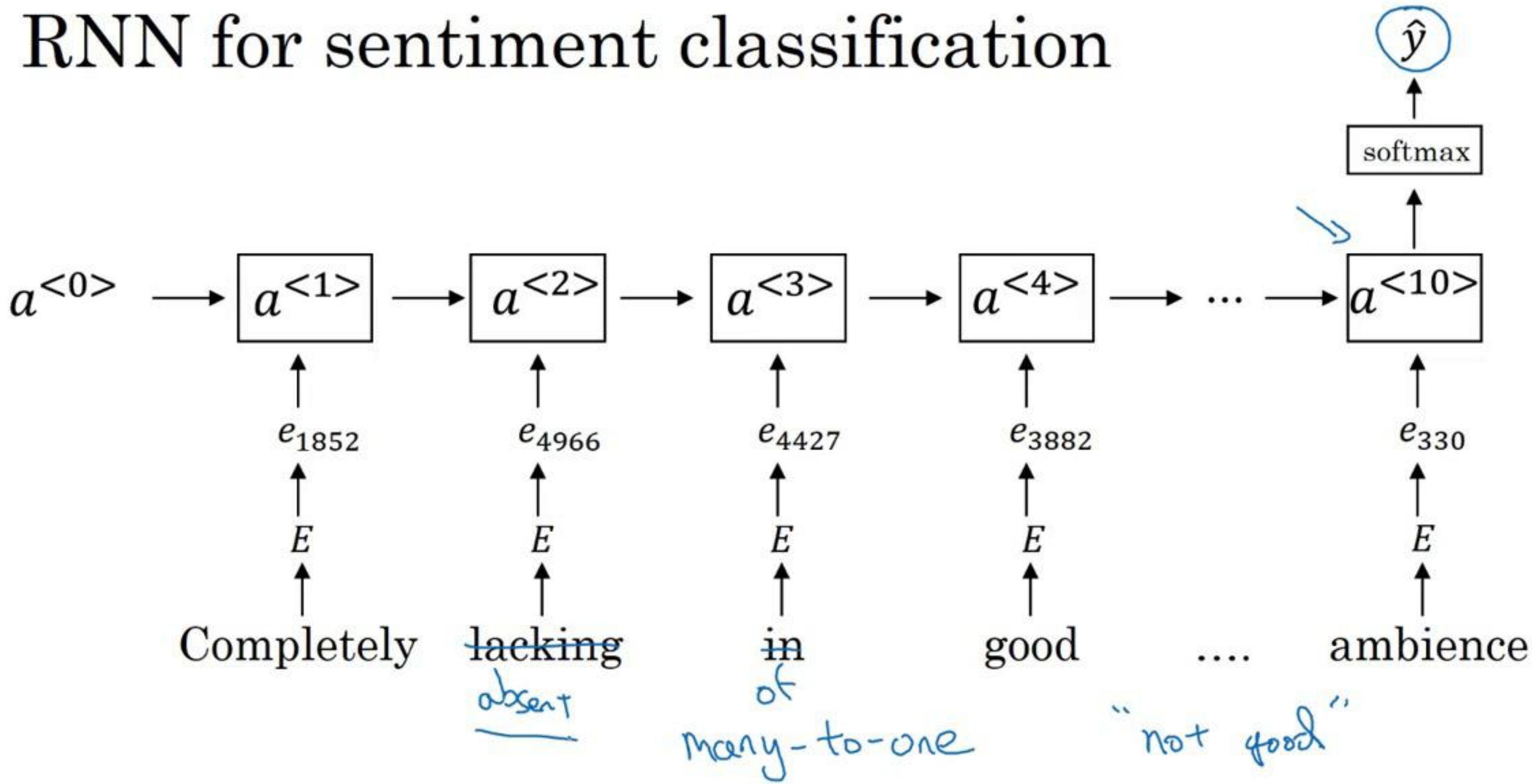
RNN for sentiment classification



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Debiasing word embeddings

The problem of bias in word embeddings

Man:Woman as King:Queen

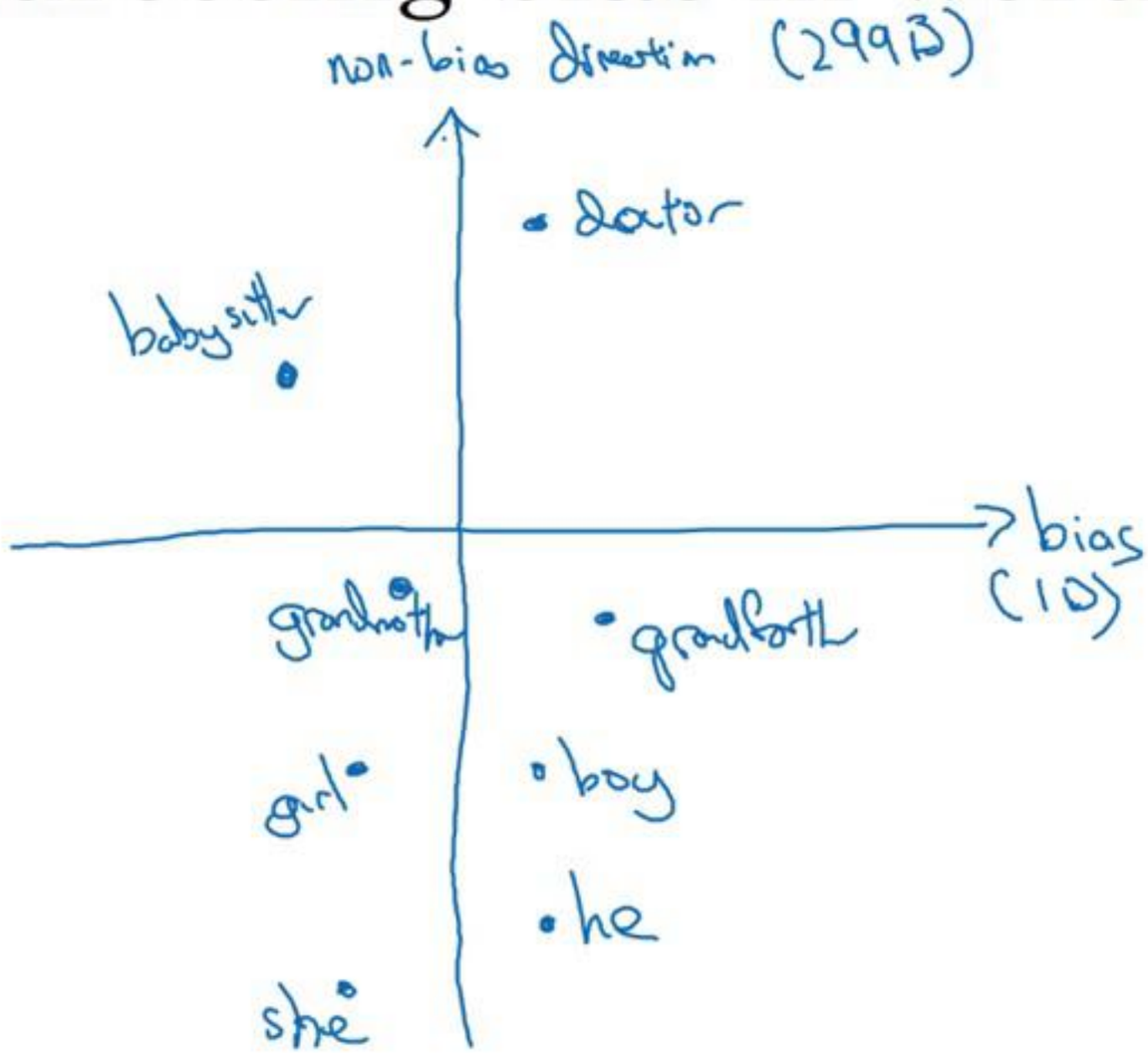
Man:Computer_Programmer as Woman:Homemaker

Father:Doctor as Mother:Nurse

Word embeddings can reflect gender, ethnicity, age, sexual orientation, and other biases of the text used to train the model.

[Bolukbasi et. al., 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings]

Addressing bias in word embeddings



1. Identify bias direction.

$$\left\{\begin{array}{l} e_{he} - e_{she} \\ e_{male} - e_{female} \\ \vdots \end{array}\right. \rightarrow \underline{\text{average}}$$

[Bolukbasi et. al., 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings]

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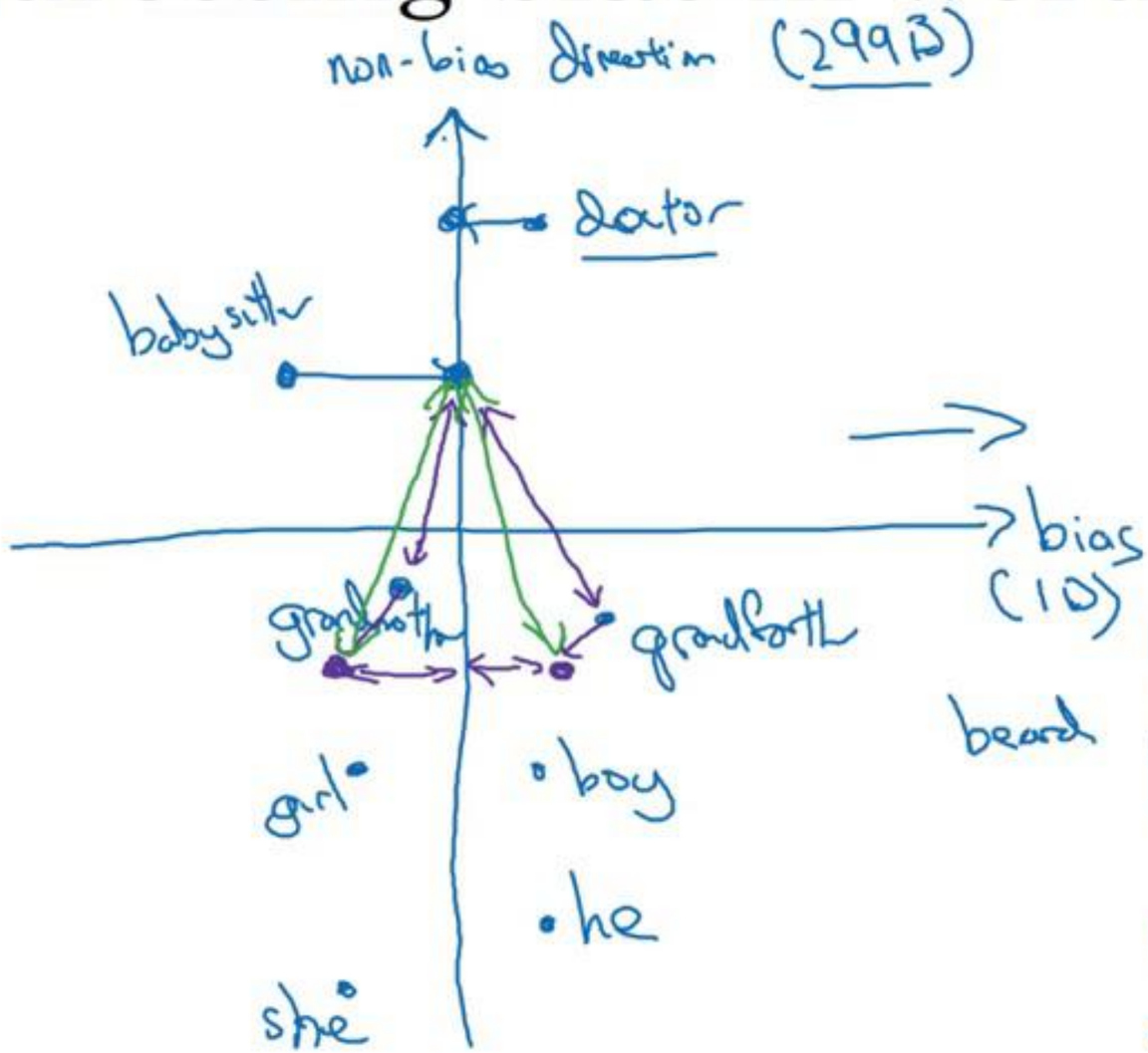
Addressing bias in word embeddings



1. Identify bias direction.

$e_{he} - e_{she}$

$e_{male} - e_{female}$

$\vdots$

average

2. Neutralize: For every word that is not definitional, project to get rid of bias.

3. Equalize pairs.

grandmother – grandfather

girl – boy

[Bolukbasi et. al., 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings]

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